Lucas Newcomb

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Education

University of Maryland, College Park — College Park, MD
B.S. Computer Engineering (intended) — *Expected May 2029*

Rochester Institute of Technology — Rochester, NY
Computer Engineering, BS/MS accelerated program; GPA: 3.98/4.00, Dean's List — *Aug. 2025 – May 2026*

- **Coursework (A in each):** Calculus I–II, University Physics I, Computer Science I–II, Digital System Design I

Technical Experience

Flight Software Responsible Engineer, Data Downlink — Oct. 2026 – Present
UMD SatDev — TERP RAPTOR — *College Park, MD*

- Own the flight software side of payload data downlink for a student CubeSat mission to the 2029 Apophis flyby.
- Building the mission's controlled data budget and proposing the onboard storage architecture, quantifying data production, buffering, retention, and transfer by mission phase against storage and downlink capacity.

Embedded Software Team Member — Aug. 2025 – May 2026
RIT Space Exploration (SPEX) — Rover Team — *Rochester, NY*

- Wrote the rover's CAN messaging library — the shared communication layer between the Teensy 4.1 controllers running the arm, chassis, and science subsystems — using dynamic dispatch to route incoming frames to registered per-message handlers, so subsystem code declares what it listens for instead of parsing the bus itself.
- Authored much of the rover's microcontroller firmware in C++ on PlatformIO; the arm and life-detection subsystems both ran on the library.
- Helped build a software simulator that mocks the rover's subsystems, letting the team exercise and test subsystem code without the physical rover.

Student Intern — July 2024 – Aug. 2024
U.S. Army Research Laboratory — *Adelphi, MD*

- Developed a prototype grant proposal application on ServiceNow, building the workflow from submission through review.
- Worked within an agile software development team alongside full-time staff.

Projects

CUBERT — Flatsat Flight Software & Ground Station | *Rust, RTIC, STM32F446, Python* — 2026 – Present

- Built end-to-end flight software for a flatsat: `no_std` Rust firmware on an STM32F446 that samples sensors on a periodic schedule, sequences behavior through a state machine, and detects and recovers from faults at runtime.
- Used RTIC for deterministic, priority-based task scheduling with no dynamic allocation, and framed downlinked telemetry over serial with COBS so packet boundaries survive a noisy link.
- Wrote the matching ground station — a Python web dashboard that decodes the live telemetry stream and renders it as time-series graphs, current values, and subsystem status indicators.

LUNA — 3D N-Body Simulator | *C++, OpenGL* — 2025 – Present

- Built a real-time gravitational N-body simulator with two interchangeable solvers — a Barnes–Hut octree at $O(n \log n)$ and direct summation at $O(n^2)$ as an accuracy reference — integrated with a symplectic leapfrog scheme that keeps orbital energy stable over long runs instead of drifting.
- Multithreaded the force computation to sustain ˜160 FPS at 10,000 bodies while remaining interactive at 100,000; renders live in OpenGL and runs headless for batch jobs.

Additional Experience

Robotics Instructor — June 2026 – Aug. 2026
Unity Active — *Northern Virginia*

- Taught robotics to school-age students with no prior technical background.

Team Member, promoted to Shift Manager — Jan. 2025 – Aug. 2026
Papa John's — *Burke, VA*

- Worked the make line and register through high-volume shifts; promoted to shift manager in summer 2026.

Technical Skills

Languages: C, C++, Rust, Python
Embedded: STM32, Teensy, CAN, I2C, SPI, UART
Systems & Tools: Linux, Git, CMake/Ninja, PlatformIO, OpenGL, clang-tidy, cppcheck, ServiceNow